

Education

- *PhD in Computing* : University of South Alabama : Fall 2017 - Spring 2020
Dissertation topic: Local Model Feature Transformations and Applications; Ryan Benton, Advisor
- *Graduate Student, Computer Science* : Rensselaer Polytechnic Institute : Fall 2012 - Spring 2014
TA for intro Python classes
- *MS Mathematics* : University of South Alabama : Fall 2006 - Fall 2008
TA for introductory-level classes
- *BS Mathematics and Statistics* : University of South Alabama : Fall 2002 - Spring 2006
Part-time department tutor and supplemental instruction leader all four years

Academia Experience

- *Assistant Professor (Lecturing) Computer Science* : University of Utah : Fall 2020 - Spring 2023
 - Designed and taught undergraduate course for Statistics for CS Majors
 - Taught introductory CS, intro to Java and CS senior capstone classes
 - Reorganized the introductory CS class to accommodate asynchronous and long-distance learning
- *Instructor Mathematics and Statistics* : University of South Alabama : Spring 2008 - Spring 2012
 - Undergraduate-level instructor for classes including Calculus, Statistics, and Pre-calculus
- *Data Science Course Mentor* : Thinkful LLC (part-time) : November 2014 - June 2015
 - Course mentor for online courses in data science and Python web programming

Applied Experience

- NOAA : *Data Scientist* : Fall 2024 -
 - Designed Hierarchical computer vision models for marine life detection and quantification
 - Architect of cloud computing infrastructure for optical data and computer vision pipelines
- USGS : *Data Scientist* : Fall 2023 - Fall 2024
 - Designed data and mapping systems for the national invasive species database
<https://siren.fort.usgs.gov/respond>
 - Designed data format and exchange systems for intersecting Ecological, Geospatial and Machine Learning data.
- National Renewable Energy Laboratory : *Internship* : Summer 2018
 - Designed a system for easily benchmarking large numbers of model families in transportation modeling problems
- Georgia Tech Research Institute : *Research Scientist II* : June 2015 - August 2017
 - Built models for city of Atlanta IOT data to determine when sensors are giving anomalous or erroneous readings
 - Built a distributed computing environment for sharing trained ML models with the wider GATech community
 - Worked with a team to build and train a deep neural network for localization and identification in images in Torch on a k40 GPU

- Set up the docker development environment for our neural network team
- Performed graph mining for the 2015 KDD cup, top performer during phase 1
- Worked with a team to develop algorithms and web frameworks to detect emerging technologies from research literature
- Created a web dashboard to display results of data mining algorithms
- Transport Foundry LLC (part-time) : *Data Scientist* : November 2014 - March 2015
 - Built a model synthesizing population transportation habits using consumer and cellular data
 - Performed QA/QC in R, in the form of graphs, statistics and reports
 - Set up from scratch AWS services for hosting and manipulating sensitive data
- CareerBuilder LLC : *Software Engineer* : June 2014 - November 2014
 - Modelled competition between jobs and applicants based on an application network with millions of nodes
 - Wrote various utilities in C# including RESTful APIs and supporting infrastructure for our relevancy and recommendations team

Publications

- Session co-organizer for TDWG 2024 Symposium "Bridging the gap: Interoperable Alien Species Data and Terminology for Successful Sharing of Management Strategies"
- CScott Brown. *Local Model Feature Transformations*. Dissertation
- CScott Brown and Ryan G. Benton. *Decision Surfaces of Localized Classifiers*. In 2020 International Joint Conference on Neural Networks (IJCNN)
- CScott Brown and Ryan G. Benton. *Local Gaussian Process Features for Clinical Sensor Time-Series*. IEEE BIBM 2019 Workshop on Machine Learning and Artificial Intelligence in Bioinformatics and Medical Informatics.
- C Scott Brown, Venu Garkiapati and Yi Hou. *A Machine-Learning Decision-Support Tool for Travel-Demand Modeling*. Presented at the Transportation Research Board annual meeting 2019.
- C. Scott Brown, William Glisson, Ryan Benton, Jordan Shropshire, Thomas Watts, and Timothy Sullivan. *Image segmentation stability: an empirical investigation*. IEEE SoutheastCon 2018.
- Al Jadda, Khalifeh, Mohammed Korayem, Camilo Ortiz, Chris Russell, David Bernal, Lamar Payson, C. Scott Brown, and Trey Grainger. *Augmenting recommendation systems using a model of semantically-related terms extracted from user behavior*. arXiv preprint arXiv:1409.2530 (2014).
- Mulekar, Madhuri S., and C. Scott Brown. *Distance and Similarity Measures*. Encyclopedia of Social Network Analysis and Mining (2014): 385-400.